

COMPARATIVE ADVANTAGE PROBLEM SET//SUGGESTED SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

- (a) In the time it takes Rakhim to wash a car he can wax 3 cars. (60 minutes/20 minutes per wax job). Therefore, Rakhim's opportunity cost of washing one car is 3 wax jobs. In the time it takes Sultan to wash one car, he can wax 2 cars. (30 minutes/15 minutes per wax job). Therefore, Sultan's opportunity cost of washing one car is 2 wax jobs. Since Sultan's opportunity cost of washing cars is lower than Rakhim's, Sultan has a comparative advantage in washing cars.
- (b) If Rakhim washes 12 cars, he gives up on waxing 4 cars. So for every car he washes, he gives up on waxing $4/12 = 1/3 \approx 0.33$ cars. Thus, Rakhim's cost of opportunity of washing one car is 0.33 wax jobs. If Sultan washes 6 cars, he gives up on waxing 3 cars. So for every car he washes, he gives up on waxing $3/6 = 1/2 = 0.5$ cars. Thus, Sultan's cost of opportunity of washing one car is 0.5 wax jobs. Thus Rakhim now has a comparative advantage in washing cars.

Problem 2

For every clutch job Akmaral gives up $4/2 = 2$ brake repairs. Therefore, Akmaral's opportunity cost of replacing one clutch is 2 sets of brakes. On the other hand, for every clutch job Abuzer gives up $6/2 = 3$ brake repairs. Therefore, Abuzer's opportunity cost of replacing one clutch is 3 brake repairs.

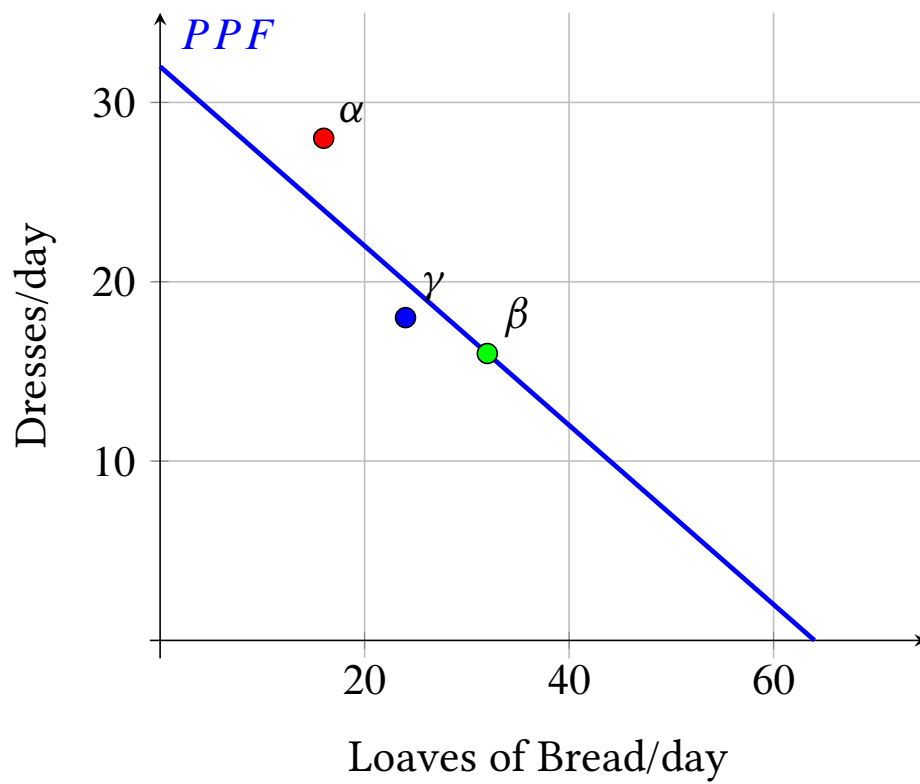
We can compute in a similar way the cost of opportunity for repairing brakes for Akmaral and Abuzer, but notice that a faster way is to just notice that this cost of opportunity is the inverse of the one we computed above. Therefore, Akmaral's cost of opportunity of repairing one set of brakes is 0.5 clutch repairs; Abuzer's cost of opportunity of repairing one set of brakes is $1/3 \approx 0.33$ clutch repairs.

Thus, Akmaral has a comparative advantage in clutch repairing, and Abuzer has a comparative advantage in brake repairing.

Finally, neither person has an absolute advantage in brake replacement since both take the same amount of time to replace a set of brakes. However, Akmaral has an absolute advantage in clutch repairs over Abuzer, since it takes her less time than Abuzer to perform a clutch repair.

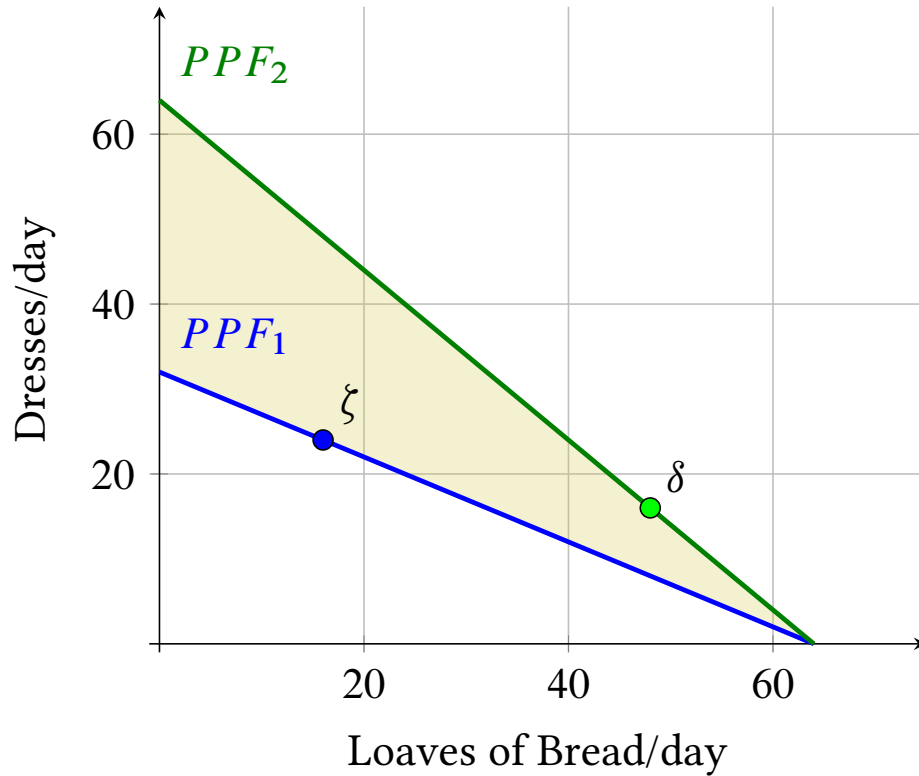
Problem 3

Let D be dresses per day and B loaves of bread per day. Then the equation representing the PPF is $D = 32 - 0.5B$. The graph of the PPF is provided below. The points $\alpha = (16, 28)$, $\beta = (32, 16)$, and $\gamma = (24, 18)$ are also shown in the graph.



Problem 4

- (a) The equation representing the new PPF is $D = 64 - B$. The graph is below.



- (b) The points $\delta = (48, 16)$, $\zeta = (16, 24)$ are also shown in the graph.
- (c) The upward rotation in Daniyar's PPF means that he is now able to produce at any point in the shaded region (in yellow) of the graph above. Note that opportunity costs have changed: for every loaf he bakes the opportunity cost in terms of dresses not sewn is now one.

Problem 5

Let N be kilograms/day of nuts and C kilograms/day of coffee. When graphing the PPF, we will put N in the horizontal axis and C in the vertical axis.

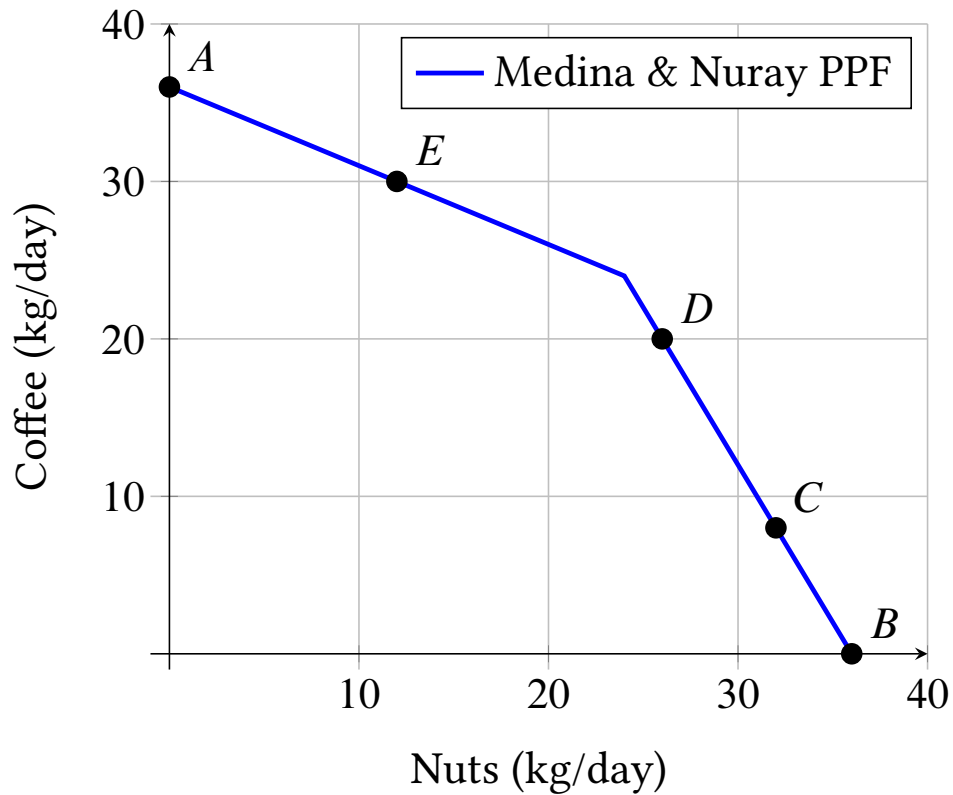
- (a) The maximum amount of coffee kilograms that they can pick in a day is: $4 \times 6 + 2 \times 6 = 36$. This corresponds to the point $A = (0, 36)$. The maximum amount of nut kilograms that they can pick in a day is: $2 \times 6 + 4 \times 6 = 36$. This corresponds to the point $B = (36, 0)$
- (b) It is instructive to construct now the following table

	Opportunity cost of 1 kg of nuts	Opportunity cost of 1 kg of coffee
Medina	2 kg of coffee	0.5 kg of nuts
Nuray	0.5 kg of coffee	2 kg of nuts

Given the information above, Medina has a comparative advantage in the production of coffee, so she should be the one picking coffee beans. It will take Medina 2 hours to pick 8 kilograms of coffee beans, and with the remaining 4 hours, she can still gather 8 kilograms of nuts. Nuray would pick 24 kilograms of nuts plus the 8 kilograms from Medina, for a total of 32 kilograms of nuts. This corresponds to the point $C = (32, 8)$.

- (c) To gather 26 kilograms of nuts per day, Nuray should specialize and work full time in the gathering of nuts (24 kilograms per day), while Medina should spend one hour per day gathering nuts (2 kilograms) and devote her remaining 5 hours to picking coffee beans, so she can pick 20 kilograms of coffee beans per day. This corresponds to the point $D = (26, 20)$.
- (d) 30 kilograms of coffee can be obtained by making Medina specialize in coffee gathering (24 kilograms per day) while Nuray spends 3 hours picking coffee beans (6 kilograms of coffee beans) and 3 hours gathering nuts (12 kilograms of nuts). This corresponds to point $E = (12, 30)$.
- (e) See the graph below. All points above are graphed and labeled in the diagram. The equation for the PPF is given by the following expression:

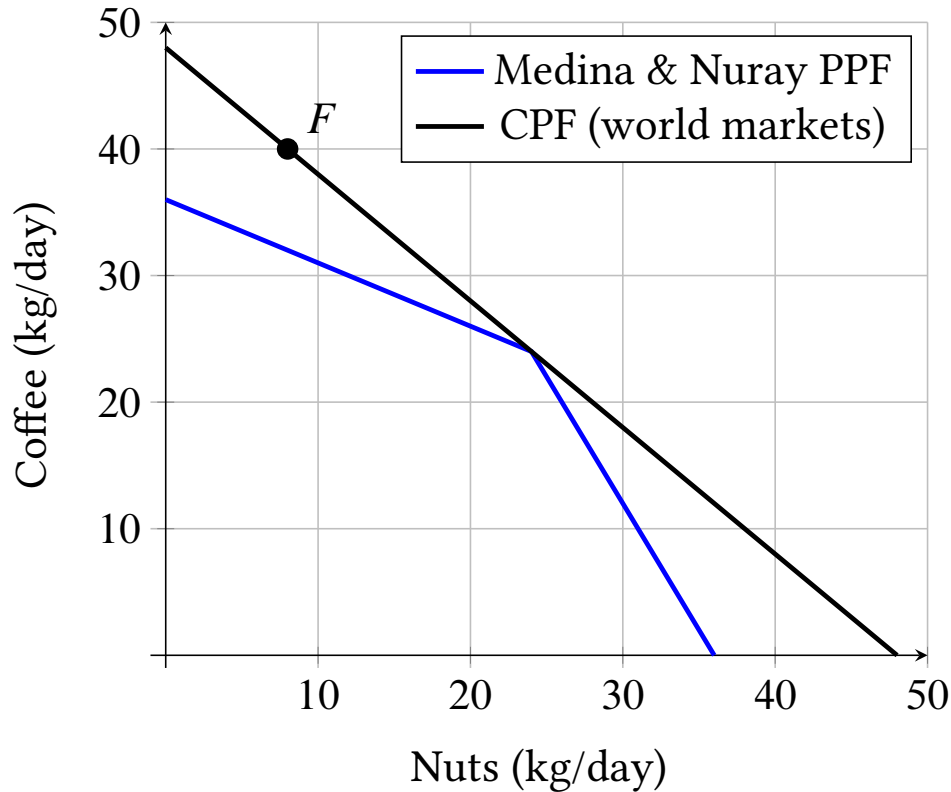
$$C = \begin{cases} 36 - 0.5N & 0 \leq N \leq 24, \\ 72 - 2N & 24 < N \leq 36. \end{cases}$$



Problem 6

- (a) By specializing completely, they can pick 24 kilograms of coffee per day and 24 kilograms of nuts per day, which is the point at which the kink occurs in the PPF diagram above. If they sell that output in the world market at the stated prices, they will receive $2 \times (24 + 24) = \$96$ per day.
- (b) With \$96 per day to spend, the maximum amount of coffee beans they could buy is 48 kilograms per day, or they could buy 48 kilograms of nuts per day. The combination of 40 kilograms and 8

kilograms of nuts would be unattainable to Medina and Nuray if they were not able to sell in the world market. However, if they buy and sell in the world market, 40 kilograms of coffee beans would cost \$80, and 8 kilograms of nuts would cost \$16, so they would have just enough money to buy this combination of goods. For an illustration, see the graph below, where $F = (8, 40)$.



- (c) The possibility to trade for Medina and Nuray in the world markets creates more possibilities of consumption. We will call this new set a *consumption possibilities set* or *CPF*. As we discussed in the previous item, by specializing, they could earn \$96 in the world markets, which they can then use to buy a maximum of 48 kilograms of coffee or 48 kilograms of nuts. This CPF can be described by the equation $C = 48 - N$. The graph of the CPF can be found in the diagram above.
- (d) If they continue to specialize, any terms of trade $\frac{p_N}{p_C}$ in the interval $[0.5, 2]$ would allow them to sell their production in the world market and still continue to benefit from a much larger consumption possibilities set compared to their original production possibilities frontier.

Problem 7

The in-home health care provided and the hair stylist are immune to outsourcing because their physical presence is required to do their job. This is not the case for computer programmers. Even though it takes considerable training and skill development to be a computer program, highly educated workers from other countries could be able to perform the same tasks compared to national workers.

Problem 8

- (a) The labor restriction in this economy is

$$l_x + l_y = 100. \quad (1)$$

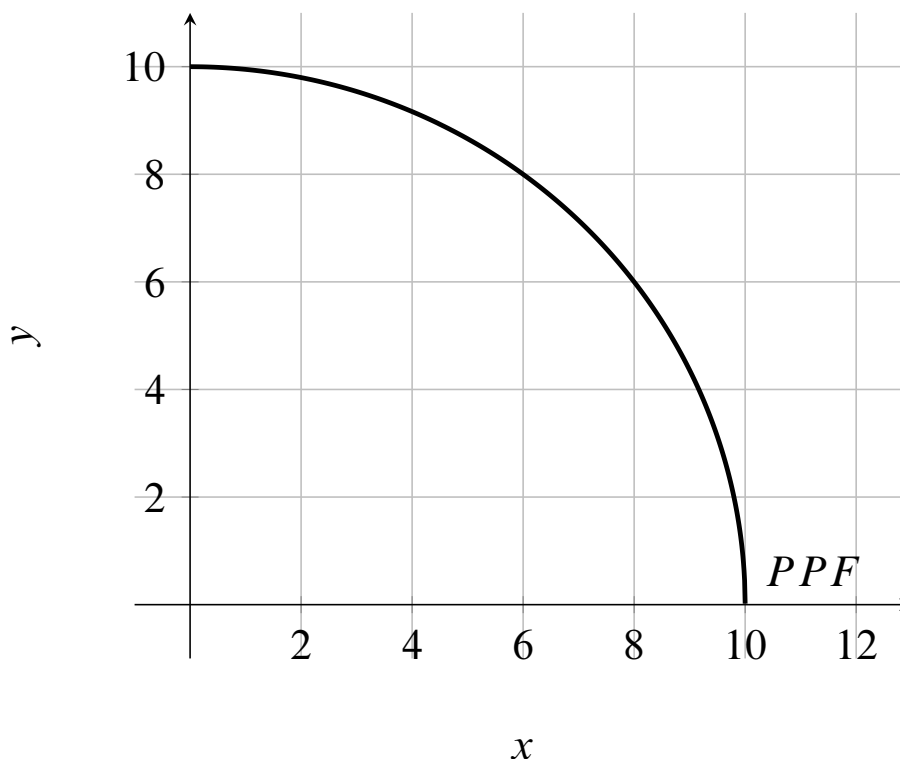
From the production functions we get that

$$l_x = x^2, \quad l_y = y^2. \quad (2)$$

Plugin the expressions in (2) into the labor restriction equation (1) yields:

$$x^2 + y^2 = 100,$$

which is the equation of a quarter circle with a radius of 10 units.



- (b) The opportunity cost of 1 unit of x in terms of units of y given up at any point (x, y) in the PPF is the slope of the PPF at point (x, y) . The interpretation is the following: if the economy is at point (x, y) , and the slope of the PPF at that point is $m(x, y)$, then if the economy tries to produce one more unit of x it will cost $|m(x, y)|$ units of y .

We can easily obtain the slope by using *implicit differentiation*. Since the PPF satisfies the relation $x^2 + y^2 = 100$, then differentiating the equation with respect to x while taking into account that y is a function of x yields:

$$2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}.$$

Consider for example the point $(6, 8)$. The slope at this point is $-\frac{6}{8} = -\frac{3}{4}$. Thus, trying to move from producing 6 units of x to 7 units of x costs $\frac{3}{4}$ units of y .